

Chatbot and Fatigued Driver: Exploring the Use of LLM-Based Voice Assistants for Driving Fatigue

Shaoshuai Huang*

School of Design and Arts, Beijing
Institute of Technology, China
3120211864@bit.edu.cn

Xuandong Zhao*

School of Design and Arts, Beijing
Institute of Technology, China
3220222098@bit.edu.cn

Dapeng Wei*

School of Design and Arts, Beijing
Institute of Technology, China
3120215958@bit.edu.cn

Xinheng Song

School of Design and Arts, Beijing
Institute of Technology, China
3120236176@bit.edu.cn

Yuanbo Sun[†]

School of Design and Arts, Beijing
Institute of Technology, China
3120215892@bit.edu.cn

ABSTRACT

This study explores the application of Large Language Model (LLM)-based voice assistants, such as ChatGPT-4, in mitigating passive driving fatigue, while aiming to enhance driving performance and safety. Employing an empirical approach with a driving simulator and the LLM-based voice assistant "Driver Mate," the study focuses on conversation complexity and frequency as independent variables, and utilizes electroencephalogram (EEG), driving simulator data, and scales for measures. The findings reveal the effectiveness of in-vehicle LLM-based voice assistants, highlighting that low-complexity, high-frequency conversation is optimal for driver alertness and acceptance, while low-complexity, low-frequency interactions significantly improve driving performance. This study innovatively investigates the role of LLM-based voice assistants in alleviating driving fatigue, offering practical suggestions for future in-vehicle systems.

CCS CONCEPTS

• Human-centered computing; • Human computer interaction (HCI); • Empirical studies in HCI;

KEYWORDS

Driving fatigue, Voice assistant, LLM, Driving performance, In-vehicle systems

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*The authors contribute equally to this paper.

[†]denotes as the corresponding author.

1 INTRODUCTION

Driving fatigue is a significant and long-standing concern for road safety, and contributing to approximately 20% of serious accidents worldwide [1]. Especially passive driving fatigue reduces driving performance by impairing attention and alertness [2], and developing driving automation makes this more likely to happen [3].

Verbal conversation with co-pilot and passengers is a common and effective method to alleviate driving fatigue [4, 5]. However, these interactions, while unpredictable and potentially increasing driving risks, can also introduce distractions and elevate cognitive workload [6].

Voice assistants based on large language models (LLMs), such as ChatGPT-4, offer the potential to improve this problem. They can engage in more intelligent natural conversations and control the form and content of their conversations [7]. While extensive research has been conducted on the utilization of LLMs for voice assistants in in-vehicle interactions [7, 8], a gap exists in the research concerning the role of LLM-based voice assistants in alleviating driving fatigue. This study seeks to fill this research gap by researching the specific ways in how LLM-based voice assistants can alleviate passive fatigue in monotonous driving through empirical research.

For this purpose, we designed an empirical study using a driving simulator, integrated with an LLM-based voice assistant, "Driver Mate," and employing a variety of measures—such as electroencephalogram (EEG), driving simulator data, and subjective rating scales—to assess drivers' alertness levels, driving performance, and system acceptance. We manipulated conversation complexity and frequency as independent variables, with a silent baseline condition for control comparison. The study involved 36 participants.

This paper contributes to the existing body of knowledge by providing empirical evidence on the effectiveness of LLM-based voice assistants in a driving context. The remainder of the paper is organized as follows: Section 2 reviews related work, Section 3 describes the method, Section 4 presents the results, followed by a discussion in Section 5, and finally, Section 6 concludes the paper with key findings and contributions.

2 RELATED WORK

2.1 Classification, effects, and measures of driving fatigue

Fatigue can be classified into active fatigue, passive fatigue, and sleep-related fatigue, which are caused by excessive task load, monotonous drive, and sleep deprivation respectively [2]. Conversation and other non-driving-related tasks (NDRTs) can be directly effective against passive fatigue, rather than other types [9, 10].

Passive fatigue can lead to alertness decrement, distraction, and worsen sleep-related fatigue, thereby reducing driving performance [2]. Passive fatigue is becoming more common and serious due to the development of advanced driver assistance systems (ADAS) [3, 10, 11], and has gradually become a concern in recent years.

Passive fatigue measurement methods include: subjective evaluation based on questionnaires (e.g., Karolinska sleepiness scale [12]), objective measurement based on biological features (e.g., EEG) and physical features (e.g., eyes state). Besides vehicle data is utilized to assess driving fatigue, as it reflects on driving performance [13].

2.2 Effects of conversation on driving fatigue and performance

Conversing with passengers while driving can reduce passive fatigue from monotonous driving, elevating alertness and workload [4, 5]. However, phone conversations tend to cause distraction, impairing driving performance [14, 15]. This difference is attributed to passengers' ability to detect traffic conditions and adjust conversation accordingly [16–18].

As for conversations with voice assistants, some studies have suggested that they can be more effective in reducing passive fatigue and improving driving performance [19–22]. Conversing with a voice assistant makes it easier to control the distracting content of the conversation than with a real person, and conversing with a voice assistant helps drivers to maintain lower psychological stress [23, 24]. There are also some counter arguments, such as Uchida et al. [6] indicate that such conversations might reduce driving performance, as drivers only attend to road during reminders.

Previous studies generally agreed that conversation can alleviate passive fatigue, but conclusions about whether conversation can enhance driving performance are not uniform. Perhaps because there are too many uncontrolled variables in the process that influence the results. Among them are the conversation complexity [6, 19, 21, 25], the duration of the conversation intervention [20, 26], conversation frequency (Continuous or interval conversations) [4, 20, 22], the ratio of speaking time to listening time [27], voice emotion [28, 29] etc. all affect the results.

2.3 LLMs-based voice AI assistant for in-vehicle interactions

Due to technical limitations, most of the previous voice assistants used in the experiments were either fully simulated dialogue or oral tasks [6, 20–22, 26, 27, 30] (e.g., pre-recorded questions played in sequence), and some studies have employed the Wizard-of-Oz (a human imitates an AI) for experiments [19, 28, 31, 32].

LLM-based assistants like ChatGPT Voice offer a more natural conversational experience than predecessors like Siri or Alexa, with

advanced comprehension and smoother, human-like interactions [7, 8, 33]. When integrated with computer vision, these assistants can adaptively support driving decisions [34]. All of which give in-vehicle voice assistants new opportunity for development. This advancement may affect driver states, potentially impacting driving performance. Large et al. [31] suggest that natural and fluid conversations with such assistants impose an acceptable cognitive load and enhance trust [35, 36]. Conversely, the immersive conversations could lead to distraction [4, 15, 24, 27].

Besides the unnatural and repetitive dialogues in previous simulation experiments can make users bored and tiresome, resulting in lower acceptance and user experience [4, 15, 27, 32]. This is also an essential issue because voice assistants can't conduct their job of influencing driving performance if the driver refuses to engage in conversation.

2.4 Aim and hypotheses

Diverging from simulated dialogues used in previous studies, this study utilizes LLM-based voice assistants for conversations to empirically investigate their role in alleviating passive fatigue during monotonous driving. Accordingly, the study proposes the following hypotheses:

- Conversation with an LLM-based voice assistant can improve the alertness and alleviate the passive fatigue during monotonous driving (H1).
- Conversations with an LLM-based voice assistant, under specific conditions of conversation complexity and frequency, can enhance driving performance during monotonous driving (H2).
- Conversations with an LLM-based voice assistant can improve drivers' user experience and acceptance (H3).

3 METHOD

3.1 Apparatus

As shown in Figure 1, a Lenovo Legion Y9000P IAH7H PC running driving simulation software displayed the interface on a triple-screen. Participants engaged in the simulation using a Playseat Sensation Pro driving seat and a Logitech G29 Driving Force steering wheel and pedals, providing a realistic driving experience. The simulation featured a straight motorway scenario, set during clear daytime conditions with no other traffic or pedestrians. An iPad equipped with the voice assistant named Driver Mate was positioned to the participant.

3.2 Participants

Thirty-six participants (19 males and 17 females) with valid driving licenses participated in the study. The average age of the participants was 24.53 years old, and the average driving experience was 4.28 years. Participants were asked to be well rested and to avoid caffeine or alcohol before the study. They received information about the study, specifically a statement about protecting the safety and privacy, and signed an informed consent form in accordance with the guidelines of the Ethics Committee of authors' university. Each participant received 50 RMB after the experiment.



Figure 1: The participant is driving in the simulator, wearing EEG cap, conversing to the voice assistant in the iPad.

3.3 Experiment design

3.3.1 Driver Mate: an LLM-based voice assistant for improving the driving fatigue and performance. In this experiment, we developed a customized GPTs named Driver Mate, a voice assistant based on ChatGPT-4. To adapt to the fatigue driving scenario, we input the following prompts for Driver Mate in advance:

- The user is in the process of driving, and your aim is to help them stay alert and focused.
- The duration of each conversation is between 30s and 60s (as recommended by Saxby et al.[16]).
- No driving, navigation or vehicle operation advice is provided.
- The language of communication is Chinese (the participants' native language).

3.3.2 Independent variables. This study utilized a 2 x 2 within-subjects design to evaluate the impact of conversation complexity and frequency using Driver Mate. Four conversations: High Complexity-High Frequency (HC-HF), High Complexity-Low Frequency (HC-LF), Low Complexity-High Frequency (LC-HF), and Low Complexity-Low Frequency (LC-LF).

Conversation complexity was based on criteria from Choudhary et al. [25]. Low-complexity involved natural conversational questions (e.g., "Where did you go on your last trip?"), while high-complexity included arithmetic problems and linguistic puzzles (e.g., "There were originally ten birds on the tree, one was shot and killed, how many birds are left on the tree?").

Frequency guidelines were derived from Mahajan et al. [22]. Low-frequency conversations initiated every two minutes, keeping a silence gap post-interaction. High-frequency conversations were nearly continuous, with only a 10-second pause for contemplation between interactions.

3.3.3 Procedure. As Figure 2, Participants began by signing an informed consent form and completing a pre-study questionnaire to collect demographic data and driving experience. This was followed by a 10-minute practice with the driving simulator, including

instructions. Then they were equipped with EEG for the study. Drivers were instructed to drive a virtual vehicle at 40 mph with cruise control and maintaining foot on the brake pedal, simulating a monotonous driving environment. This phase lasted 25 minutes to induce passive fatigue [37–39]. Then, participants underwent five 15-minute experimental rounds, interacting with Driver Mate in four modes (HC-HF, HC-LF, LC-HF, LC-LF) and a baseline non-conversation round. They filled out a post-study questionnaire and had a 5-minute silent driving recovery period [39]. The sequence of the rounds was randomized by Latin square design to mitigate sequence effects [40].

3.4 Measures

3.4.1 Alertness—for H1.

- Engagement Index (E/I) Based on EEG: The Physiological assessment of alertness was calculated through E/I (Alpha + Theta / Beta) from the frequency domain analysis during each round of driving [20, 41]. A decline in the E/I value is indicative of heightened alertness levels. Brain activity was measured using the Mitsar EEG equipment (Mitsar Co., Ltd., Russia). The EEG signals were recorded with a sampling rate of 500 Hz via 24 electrodes according to the international 10–20 system. we only report the data from the frontal lobes area (7 electrodes), which is considered as sensitive to alertness [42, 43]. To minimize noise, a 0.53Hz highpass filter and a 30Hz lowpass filter were applied, and manual inspection and Independent Component Analysis (ICA) was used.
- Karolinska Sleepiness Scale (KSS): KSS [12] was used to assess the alertness levels of participants. This subjective scale enables drivers to record their perceived level of alertness, with a range from 1 to 9.

3.4.2 Driving performance—for H2.

- Standard deviation of lane positioning (SDLP): It is estimated as the standard deviation of lateral offset of the vehicle position from center line for each driver. A higher SDLP suggests more weaving and impaired driving performance (measured in meters) [25].
- Brake response time (RT): RT means driver's response time to emergencies. Participants were instructed to brake immediately upon hearing a siren. The response time was measured from the siren's onset to the initiation of braking (measured in seconds).

3.4.3 Acceptance—for H3. Laan Questionnaire [44] was employed to evaluate driver acceptance ratings for the Driver Mate on a scale from -2 to 2, across nine attributes divided into usefulness and satisfaction.

- Usefulness: The average of items 1, 3, 5, 7, 9 (useful, good, effective, assisting, and raises alertness).
- Satisfaction: The average of items 2, 4, 6, 8 (pleasant, nice, likable, and desirable).

4 RESULT

Two-way repeated measures ANOVA were conducted on the HC-HF, HC-LF, LC-HF, and LC-LF groups to determine the significances

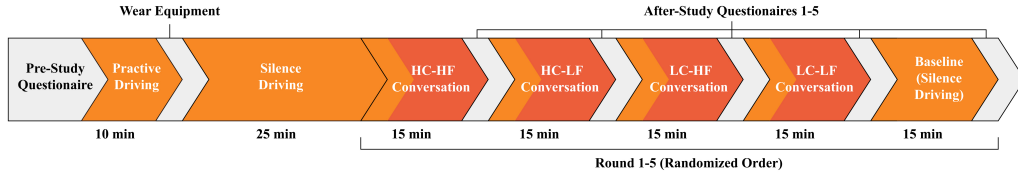
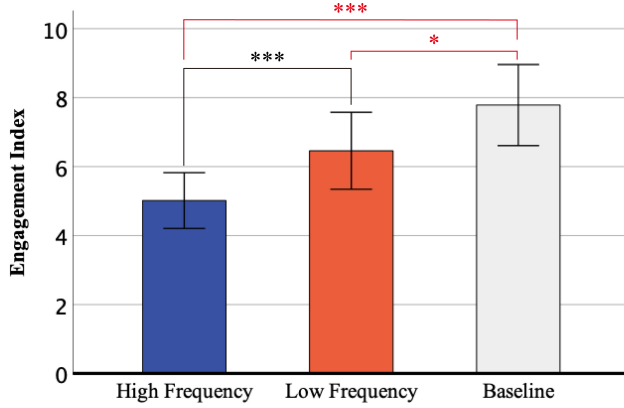


Figure 2: The experiment procedure.

Figure 3: main effect on frequency with baseline (* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$, all figures follow this annotation).

of main and interaction effects. This was followed by subsequent analysis in conjunction with baseline.

4.1 Alertness—for H1

4.1.1 Engagement Index (E/I) based on EEG. Figure 3 illustrates that E/I was significantly lower during high-frequency than in low-frequency ($F(1, 35) = 22.560$, $p < 0.001$, $\eta^2 p = 0.392$), but not significant in complexity ($F(1, 35) = 0.248$, $p = 0.622$, $\eta^2 p = 0.007$). Subsequent analysis showed that E/I decreased significantly from baseline for both high-frequency (mean = 5.017, SD = 2.387) and low-frequency (mean = 6.461, SD = 3.297), while participants in high-frequency performed better than low-frequency ($p < 0.001$). Besides, the interaction is not significant ($F(1, 35) = 0.202$, $p = 0.656$, $\eta^2 p = 0.006$).

4.1.2 Karolinska Sleepiness Scale (KSS). The analysis revealed a significant main effect of frequency on KSS ratings ($F(1, 35) = 22.008$, $p < 0.001$, $\eta^2 p = 0.386$; Figure 4 (a)), but no significant effect of complexity ($F(1, 35) = 0.064$, $p = 0.802$, $\eta^2 p = 0.002$). The Friedman test indicated a significant increase in subjective alertness from baseline in both high-frequency and low-frequency conditions ($p < 0.001$), with a significant difference between these frequencies ($p < 0.01$). Interaction was significant ($F(1, 35) = 4.628$, $p = 0.038$, $\eta^2 p = 0.117$; Figure 4 (b)). Paired comparisons showed that interaction with the Driver Mate significantly improved subjective alertness compared to baseline (all $p < 0.001$; Figure 4 (c)).

4.2 Driving performance—for H2

4.2.1 Standard deviation of lateral position (SDLP). As Figure 5, for 35 samples (1 sample data loss), high-complexity was associated with a higher SDLP significantly compared to low-complexity ($F(1, 34) = 8.070$, $p = 0.008$, $\eta^2 p = 0.192$), while no significant effects were found for complexity ($F(1, 34) = 2.364$, $p = 0.133$, $\eta^2 p = 0.065$). Comparing the high-complexity (mean = 1.977, SD = 0.698) and low-complexity (mean = 1.588, SD = 0.450) against baseline (mean = 2.057, SD = 1.109), only the low-complexity demonstrated a significant reduction in SDLP ($p = 0.019$), whereas the high-complexity did not ($p = 0.680$). And the interaction is not significant on SDLP ($F(1, 34) = 0.491$, $p = 0.488$, $\eta^2 p = 0.014$).

4.2.2 Brake response time (RT). RT revealed significant effects of both complexity ($F(1, 35) = 80.647$, $p < 0.001$, $\eta^2 p = 0.703$) and frequency ($F(1, 35) = 65.154$, $p < 0.001$, $\eta^2 p = 0.657$), with a significant interaction ($F(1, 35) = 5.472$, $p = 0.025$, $\eta^2 p = 0.139$), as shown in Figure 6 (a). Besides, Paired comparisons show only the LC-LF group demonstrated a significant reduction in RT compared to the baseline ($p = 0.002$). Even the RT of HC-HF is higher than baseline in RT, although not statistically significant.

4.3 Acceptance—for H3

The results of usefulness and satisfaction is depicted in Figure 7. High-complexity conversations were significantly less useful ($F(1, 35) = 12.689$, $p = 0.001$, $\eta^2 p = 0.266$) and less satisfying ($F(1, 35) = 10.849$, $p = 0.002$, $\eta^2 p = 0.237$) than low-complexity ones. Moreover, high-frequency conversations were rated as more useful ($F(1, 35) = 4.397$, $p = 0.043$, $\eta^2 p = 0.112$), with no significant difference in satisfaction ($F(1, 35) = 1.013$, $p = 0.321$, $\eta^2 p = 0.028$). Subsequent analysis indicated that 4 conditions of conversation scored above the baseline for both usefulness and satisfaction ratings (all $p < 0.05$). No interaction effects were significant for either usefulness ($F(1, 35) = 0.010$, $p = 0.920$, $\eta^2 p < 0.001$) or satisfaction ($F(1, 35) = 0.539$, $p = 0.468$, $\eta^2 p = 0.015$).

5 DISCUSSION

5.1 Findings and implications

5.1.1 Alertness and H1. Physiological measurements and subjective assessments consistently indicated that the use of an LLM-based voice assistant significantly mitigated passive fatigue and the decline in alertness caused by monotonous driving, thereby supporting H1. This finding aligns with previous research on the impact of conversation on passive fatigue [5, 19, 22]. High-frequency conversations were found to be more effective in enhancing alertness

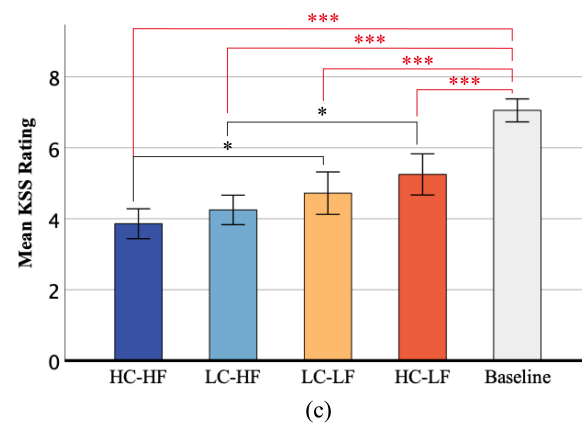
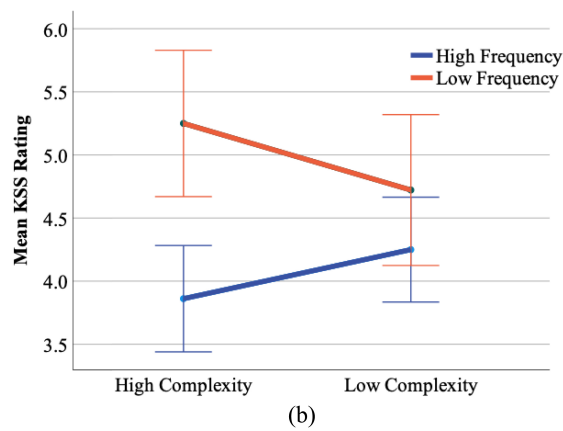
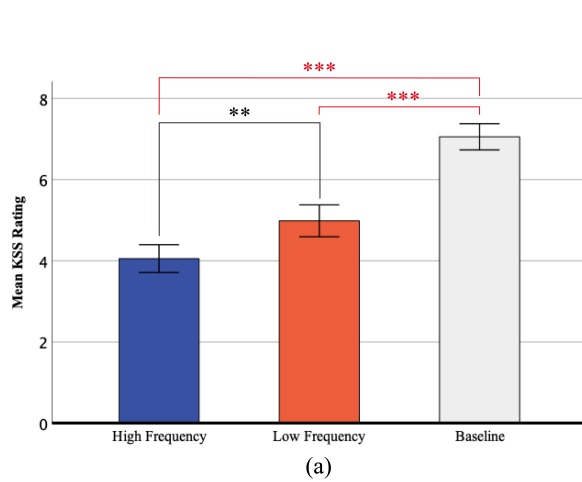


Figure 4: main effect on frequency with baseline (a), interaction effect on complexity and frequency (b), paired comparisons on above four groups and baseline (c).

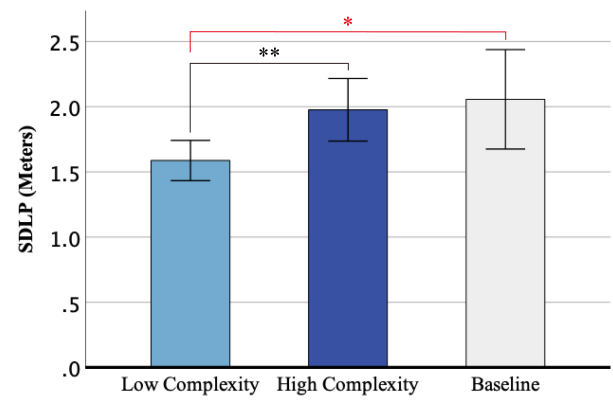


Figure 5: main effect on complexity with baseline.

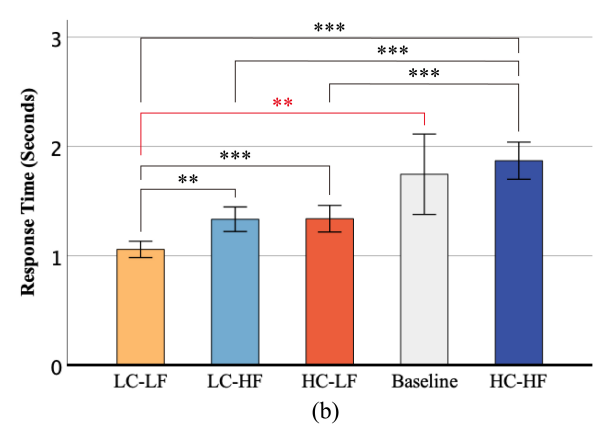
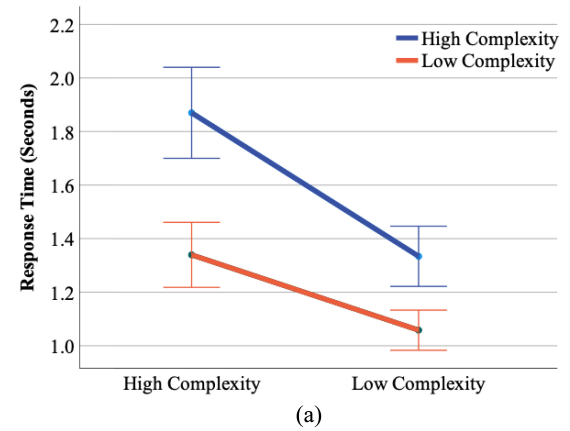


Figure 6: interaction effect on complexity and frequency (a), paired comparisons on above four groups and baseline (b).

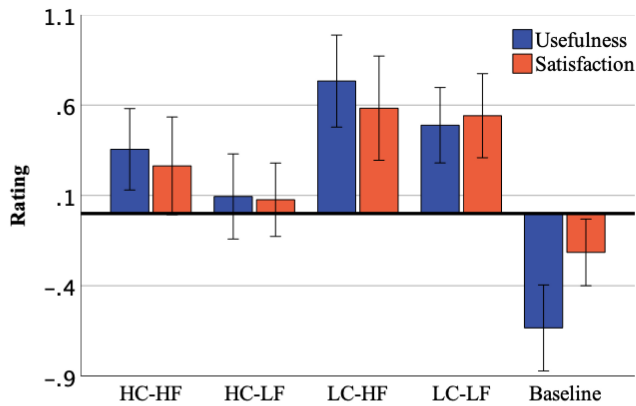


Figure 7: usefulness and satisfaction on above four groups and baseline.

compared to low-frequency conversations, but the recommendation of high-frequency conversation is contingent upon driving performance considerations. Additionally, under low-frequency, conversations of high-complexity might lead to lower subjective alertness among drivers, potentially due to the tedious nature of the topics.

5.1.2 Driving performance and H2. In terms of driving performance, analysis of SDLP and RT revealed that high-complexity conversations could impair driving performance, resulting in greater lateral movement and prolonged response times in emergency situations. Compared to baseline, significant improvements in reaction time were observed only in LC-LF. Therefore, H2 can be concluded as follows: a voice assistant of low-complexity improves lateral lane stability, while one of low-complexity and frequency effectively enhances reaction times. Given the mixed conclusions in previous studies regarding the impact of in-vehicle voice assistants on driving performance [6, 19, 21, 27, 30], it can be inferred that LLM-based voice assistants works only under specific complexity and frequency conditions.

5.1.3 Acceptance and H3. Regarding acceptance, all types of voice assistants exceeded baseline levels in terms of usefulness and satisfaction, confirming H3 that LLM-based voice assistants are highly accepted. Conversations of low-complexity scored higher in usefulness and satisfaction, indicating a preference among drivers. Moreover, high-frequency conversations were perceived as more useful, consistent with the conclusions of subjective assessments in KSS.

5.1.4 Summary and implication. In summary, in-vehicle LLM-based voice assistants under appropriate complexity and frequency can effectively alleviate driving fatigue, enhance driving performance, and are widely accepted by drivers. Therefore, the design of in-vehicle voice assistants should prioritize low-complexity conversations to improve driving performance and user acceptance. Given the considerations for alertness and acceptance, conversations of low-complexity and high-frequency also warrant consideration. It

is important to pre-set the complexity and frequency, as inappropriate settings may negatively impact driving performance. With reminders appearing only in monotonous driving segments where driver fatigue is detected, in case causes adverse effects and over-reliance on voice assistants [6, 25, 27].

5.2 Limitations and future research directions

The study's small and homogeneous sample calls for further research with varied demographics to confirm findings. Improving experimental tools, such as using a triple-screen setup or real vehicle model, could yield more realistic effects, though a single screen suffices for monotonous driving scenarios. For EEG data, only the most basic processing is applied to remove artefacts, and more precise processing could be used to improve accuracy. Afterwards, investigating unexamined aspects of LLM-based voice assistants, like conversation timing and emotional context, presents avenues for detailed future studies.

6 CONCLUSION

The results of this study validate the efficacy of LLM-based voice assistants in vehicles, demonstrating that low-complexity, high-frequency conversations are most effective in enhancing driver alertness and system acceptance, while low-complexity, low-frequency conversations notably improve driving performance. This study provides a pioneering examination of the use of LLM-based voice assistants in alleviating driver fatigue, offering valuable recommendations for future development of in-vehicle systems. These findings mark a significant progression in our understanding of voice assistant technology in the automotive sector and pave the way for continued exploration in this area.

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